# Übungsaufgaben: Gleichungen



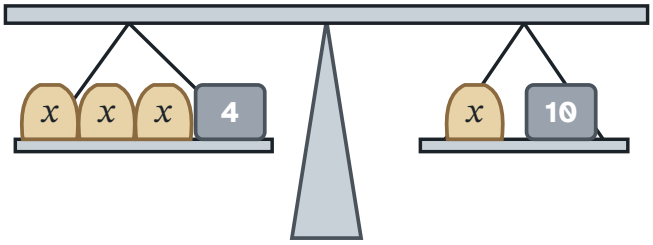
Löse die Aufgaben in deinem Heft. Die Lösungen stehen auf Seite 4.

Name: _____ Datum: _____

## ★☆☆ Leicht — Grundwissen

1. $-8x - 12 = -60$
2. $7x - 6 = 1$
3. $5x + 5 = 35$
4. $2x + 5 = -5$
5. $-x + 2 = -5$
6. $3x + 7 = 10$
7. $-2x + 9 = 19$
8. $6x + 4 = -44$
9. $-8x - 11 = 29$
10. $6x + 9 = 51$
11. $8x - 12 = -60$
12. $-2x + 1 = 7$
13. $-5x - 4 = 26$
14. $-6x + 5 = 41$
15. $-7x + 5 = -30$
16. $-9x + 12 = 57$
17. $6x + 10 = -44$
18. $-5x + 7 = -33$
19. $-3x - 4 = 8$
20. $-4x - 1 = -5$
21. $8x + 3 = -37$
22. $4x + 5 = -11$
23. $-9x - 12 = 51$
24. $6x + 7 = -5$
25. $-x + 8 = 15$
26. $-5x + 4 = -16$
27. $3x + 8 = 17$
28. $-x + 9 = 7$
29. $2x - 9 = -11$
30. $-9x + 10 = -8$
31. $8x - 10 = 62$
32. $9x + 10 = -8$
33. $3x + 6 = 33$
34. $-x + 9 = 8$
35. $-x - 4 = -11$
36. $6x - 12 = -66$
37. $-x - 8 = -2$
38. $-2x - 7 = -17$
39. $6x + 2 = -10$
40. $-x - 10 = -4$
41. $-7x - 7 = -63$
42. $-5x - 2 = 8$
43. $4x - 2 = -18$
44. $-8x + 6 = 78$
45. $6x + 1 = -35$
46. $2x + 9 = 3$
47. $-3x - 1 = -7$
48. $-9x + 5 = -40$
49. $-9x + 11 = 20$
50. $-6x - 9 = -15$
51. $-9x + 6 = -12$
52. $2x - 9 = 7$
53. $-9x - 2 = -47$
54. $5x + 4 = -1$
55. $-6x - 3 = -9$
56. $-2x - 5 = -3$
57. $6x + 3 = -21$
58. $-6x + 11 = -19$
59. $x - 1 = 7$
60. $3x + 10 = 16$
61. $6x + 10 = 46$
62. $-8x + 12 = 76$
63. $-9x - 9 = 54$
64. $-6x + 4 = -14$
65. $x + 3 = -4$
66. $-2x + 8 = -6$
67. $8x + 2 = 18$
68. $-8x + 5 = -27$
69. $-2x + 6 = 0$
70. $x + 6 = 8$
71. $2x - 9 = -27$
72. $-7x + 9 = -5$
73. $3x - 2 = 22$
74. $6x + 12 = 18$
75. $3x + 5 = -13$
76. $-9x + 12 = -33$
77. $-6x - 9 = 21$
78. $-3x + 6 = 12$
79. $-x - 11 = -6$
80. $-4x - 8 = -12$
81. $9x + 9 = 54$
82. $8x + 3 = -21$
83. $3x + 1 = 19$
84. $-8x + 1 = 9$
85. $2x + 2 = -10$
86. $6x + 1 = -5$
87. $4x - 5 = -9$

## Grafik — Gleichung an der Waage




Gewichte in kg

Die Waage ist im Gleichgewicht. Alle Säckchen wiegen gleich viel ($x$ kg).

Gleichung:

Lösung mit Rechenweg und Probe:

# Übungsaufgaben: Gleichungen



Löse die Aufgaben in deinem Heft. Die Lösungen stehen auf Seite 5.

Name: _____ Datum: _____

## ★★☆ Mittel Anwenden

**1.** $7x + 5 = x + 11$

**2.** $-2x + 6 = 3x + 16$

**3.** $2x + 9 = -x + 6$

**4.** $-3x - 8 = x + 16$

**5.** $2x - 1 = -4x - 49$

**6.** $-6x + 2 = x - 40$

**7.** $5x - 3 = -2x - 10$

**8.** $-x + 8 = x - 2$

**9.** $-5x + 9 = -x + 41$

**10.** $6x - 3 = 2x + 9$

**11.** $6x = 3x - 21$

**12.** $3x + 1 = -5x - 7$

**13.** $-x - 7 = x - 19$

**14.** $-5x + 1 = 4x + 28$

**15.** $-2x - 3 = -5x + 6$

**16.** $-4x + 1 = -3x + 9$

**17.** $3x - 1 = -5x - 41$

**18.** $-6x + 2 = -2x - 2$

**19.** $-3x + 7 = 2x - 18$

**20.** $-5x - 9 = 2x - 65$

**21.** $-5x - 3 = -x - 19$

**22.** $-3x - 8 = -2x - 2$

**23.** $-6x - 2 = 2x + 38$

**24.** $4x - 4 = 3x - 5$

**25.** $6x - 4 = 3x - 10$

**26.** $8x - 6 = 5x - 24$

**27.** $6x - 7 = x - 42$

**28.** $5x - 5 = x - 13$

**29.** $-6x - 2 = -3x - 17$

**30.** $6x + 2 = -4x + 62$

**31.** $-2x + 8 = 3x + 43$

**32.** $4x - 3 = 29$

**33.** $-x - 4 = 2x - 7$

**34.** $-5x + 7 = -3x + 17$

**35.** $x + 3 = 2x + 5$

**36.** $-5x - 7 = 2x - 28$

**37.** $-3x + 6 = x - 10$

**38.** $8x + 1 = -3x - 10$

**39.** $x + 5 = 5x - 19$

**40.** $4x + 2 = 2x + 14$

**41.** $4x - 8 = 3x - 1$

**42.** $5x + 3 = 3x - 9$

**43.** $6x - 5 = -x - 19$

**44.** $-7x - 9 = -3x + 19$

**45.** $-x - 8 = x - 14$

**46.** $4x + 1 = -5x + 64$

**47.** $-5x + 4 = 44$

**48.** $-4x - 9 = -3x - 2$

**49.** $-4x - 2 = 2x + 46$

**50.** $8x + 7 = -3x - 37$

**51.** $x + 6 = -2x + 24$

**52.** $4x - 9 = -1$

**53.** $6x + 8 = x + 48$

**54.** $4x + 2 = 3x - 1$

**55.** $8x + 7 = x + 21$

**56.** $8x - 2 = 14$

**57.** $5x + 9 = -4x$

**58.** $3x + 4 = 2x + 1$

**59.** $-6x - 6 = -5x - 7$

**60.** $-4x - 3 = 4x + 13$

**61.** $8x - 7 = x + 7$

**62.** $2x + 1 = -3x - 24$

**63.** $5x = x - 28$

**64.** $7x - 6 = x + 30$

**65.** $5x - 4 = 26$

**66.** $3x + 9 = -x - 23$

**67.** $4x + 1 = -4x - 23$

**68.** $-5x + 1 = x - 23$

**69.** $-3x - 4 = -2x - 11$

**70.** $8x - 8 = -5x + 96$

**71.** $-3x + 1 = -x - 7$

**72.** $5x - 4 = x + 8$

**73.** $6x + 9 = 2x - 11$

**74.** $-3x - 9 = 0$

**75.** $-5x + 6 = 4x - 12$

**76.** $-7x + 7 = -x + 55$

**77.** $3x - 2 = -2x - 7$

**78.** $-2x - 1 = -3x - 3$

**79.** $x - 6 = -3x - 18$

**80.** $8x - 9 = -4x + 15$

**81.** $8x + 7 = -4x - 5$

**82.** $-5x + 8 = 3x - 24$

**83.** $7x - 8 = -x - 40$

**84.** $-5x - 1 = x + 23$

**85.** $-6x - 8 = x - 29$

**86.** $8x + 1 = -5x - 103$

**87.** $7x = 56$

**88.** $8x + 2 = -x - 61$

**89.** $-3x + 6 = -x - 4$

**90.** $7x - 5 = 16$

**91.** $3x - 2 = x + 4$

**92.** $-5x + 1 = -3x + 5$

**93.** $-x + 1 = 5x - 35$

**94.** $-4x + 2 = 5x + 65$

**95.** $6x + 3 = 3x + 18$

**96.** $4x - 4 = 3x - 7$

**97.** $5x + 4 = -2x + 32$

**98.** $-4x - 1 = 2x - 7$

**99.** $-x + 6 = 5x + 24$

**100.** $-5x + 6 = 26$

**101** $8x = -4x - 48$

**102** $x - 7 = -5x - 1$

**103.** $-6x + 3 = -4x - 1$

**104.** $-7x + 4 = -2x - 36$

**105** $4x - 6 = -4x + 10$

**106** $2x + 3 = 3x - 5$

**107** $2x - 8 = -3x + 17$

**108** $5x - 1 = x + 23$

**109.** $-6x - 8 = 3x - 71$

**110.** $-4x - 2 = 4x + 14$

**111.** $-7x - 9 = 2x + 36$

**112.** $-2x = 5x - 28$

**113.** $-3x + 4 = 2x - 36$

**114.** $-7x + 1 = -5x - 11$

**115** $3x + 8 = 4x + 12$

**116** $x + 6 = 5x + 34$

**117** $8x + 8 = -5x - 44$

# Übungsaufgaben: Gleichungen



Löse die Aufgaben in deinem Heft. Die Lösungen stehen auf Seite 6.

Name: _____ Datum: _____

## ★★★ Schwer

Knobeln und Transfer

1. $-4(x+5)=2x-35$
2. $\frac{x}{4}-4=1$
3. $(x+2)(x-5)=(x+4)(x+2)-27$
4. $3(2x+1)=-x+10$
5. $\frac{x}{4}-4=-7$
6. $(x+2)(x+4)=(x+3)(x-3)-19$
7. $4(2x-5)=-4x-92$
8. $\frac{x}{5}+6=1$
9. $(x+3)(x-2)=(x+1)(x-2)-2$
10. $6(x+6)=-2x+52$
11. $\frac{x}{3}-6=-1$
12. $(x+4)(x-1)=(x+1)(x-2)+6$
13. $-2(x+5)=3x-30$
14. $\frac{x}{3}-2=-5$
15. $(x+4)(x-5)=(x+2)(x-5)-2$
16. $-4(2x+2)=x+24$
17. $\frac{x}{4}+1=0$
18. $(x+2)(x-5)=(x+4)(x+2)$
19. $-4(3x-6)=3x-21$
20. $\frac{x}{4}-3=-8$
21. $(x+4)(x+4)=(x+4)(x-4)+16$
22. $-3(x-6)=3x-6$
23. $\frac{x}{4}-5=0$
24. $(x+4)(x-5)=(x+1)(x-1)-16$
25. $-5(2x-3)=x-40$
26. $\frac{x}{2}-4=-2$
27. $(x+4)(x+1)=(x+2)(x-2)-2$
28. $-1(x-1)=x-9$
29. $\frac{x}{3}-1=-5$
30. $(x+1)(x+2)=(x+3)(x+2)-12$
31. $3(3x-2)=2x+15$
32. $\frac{x}{5}+2=7$
33. $(x+4)(x-5)=(x+1)(x+5)-32$
34. $6(2x+4)=3x+20$
35. $\frac{x}{2}-3=-9$
36. $(x+1)(x-5)=(x+1)(x+2)-28$
37. $-3(3x-5)=-39$
38. $\frac{x}{4}-1=3$
39. $(x+3)(x-5)=(x+4)(x-5)+7$
40. $2(x-6)=-26$
41. $\frac{x}{5}-4=-2$
42. $(x+4)(x+1)=(x+4)(x-4)+15$
43. $-1(x+3)=-3x+5$
44. $\frac{x}{3}+5=1$
45. $(x+3)(x+4)=(x+3)(x-3)$
46. $6(x-4)=-x-45$
47. $\frac{x}{3}-1=-6$
48. $(x+1)(x+2)=(x+2)(x+2)+1$
49. $-1(3x-6)=-3$
50. $\frac{x}{4}+4=5$
51. $(x+1)(x+1)=(x+2)(x-5)+31$
52. $6(3x-4)=2x-88$
53. $\frac{x}{4}-4=-5$
54. $(x+2)(x-2)=(x+1)(x-3)-13$
55. $-3(3x-4)=-2x-30$
56. $\frac{x}{5}+5=8$
57. $(x+2)(x+4)=(x+1)(x+6)+6$
58. $2(3x+2)=-4x-36$
59. $\frac{x}{3}-6=-12$
60. $(x+2)(x-2)=(x+4)(x+5)-69$
61. $5(x-1)=4x-10$
62. $\frac{x}{3}+6=12$
63. $(x+4)(x+3)=(x+4)(x-2)+35$
64. $-5(2x+2)=-x+26$
65. $\frac{x}{3}-3=-2$
66. $(x+2)(x+5)=(x+2)(x+4)+7$
67. $-1(2x+5)=2x+9$
68. $\frac{x}{2}+4=10$
69. $(x+3)(x-1)=(x+2)(x-5)+12$
70. $3(x-1)=-2x+2$
71. $\frac{x}{4}-5=-1$
72. $(x+4)(x-2)=(x+2)(x-2)$
73. $1(x+2)=-4x+7$
74. $\frac{x}{4}+1=4$
75. $(x+2)(x-4)=(x+2)(x-1)-21$
76. $-5(x-4)=3x+12$
77. $\frac{x}{5}-6=-1$
78. $(x+1)(x+3)=(x+2)(x+5)-19$
79. $2(2x-2)=-2x+14$
80. $\frac{x}{5}+5=0$
81. $(x+3)(x+1)=(x+1)(x-5)+24$
82. $5(3x-3)=-3x+129$
83. $\frac{x}{5}+6=11$
84. $(x+1)(x-4)=(x+4)(x+2)-30$
85. $-2(3x+1)=4x-27$
86. $\frac{x}{4}+6=8$
87. $(x+3)(x+1)=(x+2)(x+1)+3$
88. $5(x+2)=2x+22$
89. $(x+4)(x-1)=(x+1)(x-3)-31$
90. $-3(2x+4)=2x-4$
91. $\frac{x}{2}-6=-11$
92. $(x+1)(x-2)=(x+3)(x+2)-38$
93. $-1(3x-6)=3x-42$
94. $\frac{x}{4}+2=-1$
95. $(x+1)(x+3)=(x+3)(x-1)+4$
96. $-3(3x-1)=2x+36$
97. $\frac{x}{3}+4=6$
98. $(x+1)(x+5)=(x+4)(x-3)+32$

# Lösungen: Gleichungen

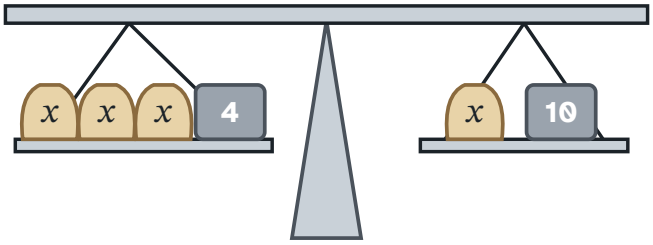


Lösungen zu Seite 1 (Leicht). Vergleiche mit deinen Ergebnissen.

## ★☆☆ Leicht — Grundwissen

1. $-8x = -48 \Rightarrow x = 6$
2. $7x = 7 \Rightarrow x = 1$
3. $5x = 30 \Rightarrow x = 6$
4. $2x = -10 \Rightarrow x = -5$
5. $-x = -7 \Rightarrow x = 7$
6. $3x = 3 \Rightarrow x = 1$
7. $-2x = 10 \Rightarrow x = -5$
8. $6x = -48 \Rightarrow x = -8$
9. $-8x = 40 \Rightarrow x = -5$
10. $6x = 42 \Rightarrow x = 7$
11. $8x = -48 \Rightarrow x = -6$
12. $-2x = 6 \Rightarrow x = -3$
13. $-5x = 30 \Rightarrow x = -6$
14. $-6x = 36 \Rightarrow x = -6$
15. $-7x = -35 \Rightarrow x = 5$
16. $-9x = 45 \Rightarrow x = -5$
17. $6x = -54 \Rightarrow x = -9$
18. $-5x = -40 \Rightarrow x = 8$
19. $-3x = 12 \Rightarrow x = -4$
20. $-4x = -4 \Rightarrow x = 1$
21. $8x = -40 \Rightarrow x = -5$
22. $4x = -16 \Rightarrow x = -4$
23. $-9x = 63 \Rightarrow x = -7$
24. $6x = -12 \Rightarrow x = -2$
25. $-x = 7 \Rightarrow x = -7$
26. $-5x = -20 \Rightarrow x = 4$
27. $3x = 9 \Rightarrow x = 3$
28. $-x = -2 \Rightarrow x = 2$
29. $2x = -2 \Rightarrow x = -1$
30. $-9x = -18 \Rightarrow x = 2$
31. $8x = 72 \Rightarrow x = 9$
32. $9x = -18 \Rightarrow x = -2$
33. $3x = 27 \Rightarrow x = 9$
34. $-x = -1 \Rightarrow x = 1$
35. $-x = -7 \Rightarrow x = 7$
36. $6x = -54 \Rightarrow x = -9$
37. $-x = 6 \Rightarrow x = -6$
38. $-2x = -10 \Rightarrow x = 5$
39. $6x = -12 \Rightarrow x = -2$
40. $-x = 6 \Rightarrow x = -6$
41. $-7x = -56 \Rightarrow x = 8$
42. $-5x = 10 \Rightarrow x = -2$
43. $4x = -16 \Rightarrow x = -4$
44. $-8x = 72 \Rightarrow x = -9$
45. $6x = -36 \Rightarrow x = -6$
46. $2x = -6 \Rightarrow x = -3$
47. $-3x = -6 \Rightarrow x = 2$
48. $-9x = -45 \Rightarrow x = 5$
49. $-9x = 9 \Rightarrow x = -1$
50. $-6x = -6 \Rightarrow x = 1$
51. $-9x = -18 \Rightarrow x = 2$
52. $2x = 16 \Rightarrow x = 8$
53. $-9x = -45 \Rightarrow x = 5$
54. $5x = -5 \Rightarrow x = -1$
55. $-6x = -6 \Rightarrow x = 1$
56. $-2x = 2 \Rightarrow x = -1$
57. $6x = -24 \Rightarrow x = -4$
58. $-6x = -30 \Rightarrow x = 5$
59. $x = 8 \Rightarrow x = 8$
60. $3x = 6 \Rightarrow x = 2$
61. $6x = 36 \Rightarrow x = 6$
62. $-8x = 64 \Rightarrow x = -8$
63. $-9x = 63 \Rightarrow x = -7$
64. $-6x = -18 \Rightarrow x = 3$
65. $x = -7 \Rightarrow x = -7$
66. $-2x = -14 \Rightarrow x = 7$
67. $8x = 16 \Rightarrow x = 2$
68. $-8x = -32 \Rightarrow x = 4$
69. $-2x = -6 \Rightarrow x = 3$
70. $x = 2 \Rightarrow x = 2$
71. $2x = -18 \Rightarrow x = -9$
72. $-7x = -14 \Rightarrow x = 2$
73. $3x = 24 \Rightarrow x = 8$
74. $6x = 6 \Rightarrow x = 1$
75. $3x = -18 \Rightarrow x = -6$
76. $-9x = -45 \Rightarrow x = 5$
77. $-6x = 30 \Rightarrow x = -5$
78. $-3x = 6 \Rightarrow x = -2$
79. $-x = 5 \Rightarrow x = -5$
80. $-4x = -4 \Rightarrow x = 1$
81. $9x = 45 \Rightarrow x = 5$
82. $8x = -24 \Rightarrow x = -3$
83. $3x = 18 \Rightarrow x = 6$
84. $-8x = 8 \Rightarrow x = -1$
85. $2x = -12 \Rightarrow x = -6$
86. $6x = -6 \Rightarrow x = -1$
87. $4x = -4 \Rightarrow x = -1$

## Grafik — Gleichung an der Waage




Gewichte in kg

$3x + 4 = x + 10$ | $- x$ → $2x + 4 = 10$ | $- 4$ → $2x = 6$ | $: 2$ → $x = 3$

Probe: $3 \cdot 3 + 4 = 13$ und $3 + 10 = 13$ ✓. Ein Säckchen wiegt **3 kg**.

# Lösungen: Gleichungen



Lösungen zu Seite 2 (Mittel). Vergleiche mit deinen Ergebnissen.

## ★★☆ Mittel Anwenden

1. $6x = 6 \Rightarrow x = 1$
2. $-5x = 10 \Rightarrow x = -2$
3. $3x = -3 \Rightarrow x = -1$
4. $-4x = 24 \Rightarrow x = -6$
5. $6x = -48 \Rightarrow x = -8$
6. $-7x = -42 \Rightarrow x = 6$
7. $7x = -7 \Rightarrow x = -1$
8. $-2x = -10 \Rightarrow x = 5$
9. $-4x = 32 \Rightarrow x = -8$
10. $4x = 12 \Rightarrow x = 3$
11. $3x = -21 \Rightarrow x = -7$
12. $8x = -8 \Rightarrow x = -1$
13. $-2x = -12 \Rightarrow x = 6$
14. $-9x = 27 \Rightarrow x = -3$
15. $3x = 9 \Rightarrow x = 3$
16. $-x = 8 \Rightarrow x = -8$
17. $8x = -40 \Rightarrow x = -5$
18. $-4x = -4 \Rightarrow x = 1$
19. $-5x = -25 \Rightarrow x = 5$
20. $-7x = -56 \Rightarrow x = 8$
21. $-4x = -16 \Rightarrow x = 4$
22. $-x = 6 \Rightarrow x = -6$
23. $-8x = 40 \Rightarrow x = -5$
24. $x = -1 \Rightarrow x = -1$
25. $3x = -6 \Rightarrow x = -2$
26. $3x = -18 \Rightarrow x = -6$
27. $5x = -35 \Rightarrow x = -7$
28. $4x = -8 \Rightarrow x = -2$
29. $-3x = -15 \Rightarrow x = 5$
30. $10x = 60 \Rightarrow x = 6$
31. $-5x = 35 \Rightarrow x = -7$
32. $4x = 32 \Rightarrow x = 8$
33. $-3x = -3 \Rightarrow x = 1$
34. $-2x = 10 \Rightarrow x = -5$
35. $-x = 2 \Rightarrow x = -2$
36. $-7x = -21 \Rightarrow x = 3$
37. $-4x = -16 \Rightarrow x = 4$
38. $11x = -11 \Rightarrow x = -1$
39. $-4x = -24 \Rightarrow x = 6$
40. $2x = 12 \Rightarrow x = 6$
41. $x = 7 \Rightarrow x = 7$
42. $2x = -12 \Rightarrow x = -6$
43. $7x = -14 \Rightarrow x = -2$
44. $-4x = 28 \Rightarrow x = -7$
45. $-2x = -6 \Rightarrow x = 3$
46. $9x = 63 \Rightarrow x = 7$
47. $-5x = 40 \Rightarrow x = -8$
48. $-x = 7 \Rightarrow x = -7$
49. $-6x = 48 \Rightarrow x = -8$
50. $11x = -44 \Rightarrow x = -4$
51. $3x = 18 \Rightarrow x = 6$
52. $4x = 8 \Rightarrow x = 2$
53. $5x = 40 \Rightarrow x = 8$
54. $x = -3 \Rightarrow x = -3$
55. $7x = 14 \Rightarrow x = 2$
56. $8x = 16 \Rightarrow x = 2$
57. $9x = -9 \Rightarrow x = -1$
58. $x = -3 \Rightarrow x = -3$
59. $-x = -1 \Rightarrow x = 1$
60. $-8x = 16 \Rightarrow x = -2$
61. $7x = 14 \Rightarrow x = 2$
62. $5x = -25 \Rightarrow x = -5$
63. $4x = -28 \Rightarrow x = -7$
64. $6x = 36 \Rightarrow x = 6$
65. $5x = 30 \Rightarrow x = 6$
66. $4x = -32 \Rightarrow x = -8$
67. $8x = -24 \Rightarrow x = -3$
68. $-6x = -24 \Rightarrow x = 4$
69. $-x = -7 \Rightarrow x = 7$
70. $13x = 104 \Rightarrow x = 8$
71. $-2x = -8 \Rightarrow x = 4$
72. $4x = 12 \Rightarrow x = 3$
73. $4x = -20 \Rightarrow x = -5$
74. $-3x = 9 \Rightarrow x = -3$
75. $-9x = -18 \Rightarrow x = 2$
76. $-6x = 48 \Rightarrow x = -8$
77. $5x = -5 \Rightarrow x = -1$
78. $x = -2 \Rightarrow x = -2$
79. $4x = -12 \Rightarrow x = -3$
80. $12x = 24 \Rightarrow x = 2$
81. $12x = -12 \Rightarrow x = -1$
82. $-8x = -32 \Rightarrow x = 4$
83. $8x = -32 \Rightarrow x = -4$
84. $-6x = 24 \Rightarrow x = -4$
85. $-7x = -21 \Rightarrow x = 3$
86. $13x = -104 \Rightarrow x = -8$
87. $7x = 56 \Rightarrow x = 8$
88. $9x = -63 \Rightarrow x = -7$
89. $-2x = -10 \Rightarrow x = 5$
90. $7x = 21 \Rightarrow x = 3$
91. $2x = 6 \Rightarrow x = 3$
92. $-2x = 4 \Rightarrow x = -2$
93. $-6x = -36 \Rightarrow x = 6$
94. $-9x = 63 \Rightarrow x = -7$
95. $3x = 15 \Rightarrow x = 5$
96. $x = -3 \Rightarrow x = -3$
97. $7x = 28 \Rightarrow x = 4$
98. $-6x = -6 \Rightarrow x = 1$
99. $-6x = 18 \Rightarrow x = -3$
100. $-5x = 20 \Rightarrow x = -4$
101. $12x = -48 \Rightarrow x = -4$
102. $6x = 6 \Rightarrow x = 1$
103. $-2x = -4 \Rightarrow x = 2$
104. $-5x = -40 \Rightarrow x = 8$
105. $8x = 16 \Rightarrow x = 2$
106. $-x = -8 \Rightarrow x = 8$
107. $5x = 25 \Rightarrow x = 5$
108. $4x = 24 \Rightarrow x = 6$
109. $-9x = -63 \Rightarrow x = 7$
110. $-8x = 16 \Rightarrow x = -2$
111. $-9x = 45 \Rightarrow x = -5$
112. $-7x = -28 \Rightarrow x = 4$
113. $-5x = -40 \Rightarrow x = 8$
114. $-2x = -12 \Rightarrow x = 6$
115. $-x = 4 \Rightarrow x = -4$
116. $-4x = 28 \Rightarrow x = -7$
117. $13x = -52 \Rightarrow x = -4$

# Lösungen: Gleichungen



Lösungen zu Seite 3 (Schwer). Vergleiche mit deinen Ergebnissen.

## ★★★ Schwer — Knobeln und Transfer

1. $-4x - 20 = 2x - 35 \Rightarrow x = \frac{5}{2}$
2. $\frac{x}{4} = 5 \Rightarrow x = 20$
3. $x^2$ fällt weg: $-3x - 10 = 6x - 19 \Rightarrow x = 1$
4. $6x + 3 = -x + 10 \Rightarrow x = 1$
5. $\frac{x}{4} = -3 \Rightarrow x = -12$
6. $x^2$ fällt weg: $6x + 8 = -28 \Rightarrow x = -6$
7. $8x - 20 = -4x - 92 \Rightarrow x = -6$
8. $\frac{x}{5} = -5 \Rightarrow x = -25$
9. $x^2$ fällt weg: $x - 6 = -x - 4 \Rightarrow x = 1$
10. $6x + 36 = -2x + 52 \Rightarrow x = 2$
11. $\frac{x}{3} = 5 \Rightarrow x = 15$
12. $x^2$ fällt weg: $3x - 4 = -x + 4 \Rightarrow x = 2$
13. $-2x - 10 = 3x - 30 \Rightarrow x = 4$
14. $\frac{x}{3} = -3 \Rightarrow x = -9$
15. $x^2$ fällt weg: $-x - 20 = -3x - 12 \Rightarrow x = 4$
16. $-8x - 8 = x + 24 \Rightarrow x = -\frac{32}{9}$
17. $\frac{x}{4} = -1 \Rightarrow x = -4$
18. $x^2$ fällt weg: $-3x - 10 = 6x + 8 \Rightarrow x = -2$
19. $-12x + 24 = 3x - 21 \Rightarrow x = 3$
20. $\frac{x}{4} = -5 \Rightarrow x = -20$
21. $x^2$ fällt weg: $8x + 16 = 0 \Rightarrow x = -2$
22. $-3x + 18 = 3x - 6 \Rightarrow x = 4$
23. $\frac{x}{4} = 5 \Rightarrow x = 20$
24. $x^2$ fällt weg: $-x - 20 = -17 \Rightarrow x = -3$
25. $-10x + 15 = x - 40 \Rightarrow x = 5$
26. $\frac{x}{2} = 2 \Rightarrow x = 4$
27. $x^2$ fällt weg: $5x + 4 = -6 \Rightarrow x = -2$
28. $-x + 1 = x - 9 \Rightarrow x = 5$
29. $\frac{x}{3} = -4 \Rightarrow x = -12$
30. $x^2$ fällt weg: $3x + 2 = 5x - 6 \Rightarrow x = 4$
31. $9x - 6 = 2x + 15 \Rightarrow x = 3$
32. $\frac{x}{5} = 5 \Rightarrow x = 25$
33. $x^2$ fällt weg: $-x - 20 = 6x - 27 \Rightarrow x = 1$
34. $12x + 24 = 3x + 20 \Rightarrow x = -\frac{4}{9}$
35. $\frac{x}{2} = -6 \Rightarrow x = -12$
36. $x^2$ fällt weg: $-4x - 5 = 3x - 26 \Rightarrow x = 3$
37. $-9x + 15 = -39 \Rightarrow x = 6$
38. $\frac{x}{4} = 4 \Rightarrow x = 16$
39. $x^2$ fällt weg: $-2x - 15 = -x - 13 \Rightarrow x = -2$
40. $2x - 12 = -26 \Rightarrow x = -7$
41. $\frac{x}{5} = 2 \Rightarrow x = 10$
42. $x^2$ fällt weg: $5x + 4 = -1 \Rightarrow x = -1$
43. $-x - 3 = -3x + 5 \Rightarrow x = 4$
44. $\frac{x}{3} = -4 \Rightarrow x = -12$
45. $x^2$ fällt weg: $7x + 12 = -9 \Rightarrow x = -3$
46. $6x - 24 = -x - 45 \Rightarrow x = -3$
47. $\frac{x}{3} = -5 \Rightarrow x = -15$
48. $x^2$ fällt weg: $3x + 2 = 4x + 5 \Rightarrow x = -3$
49. $-3x + 6 = -3 \Rightarrow x = 3$
50. $\frac{x}{4} = 1 \Rightarrow x = 4$
51. $x^2$ fällt weg: $2x + 1 = -3x + 21 \Rightarrow x = 4$
52. $18x - 24 = 2x - 88 \Rightarrow x = -4$
53. $\frac{x}{4} = -1 \Rightarrow x = -4$
54. $x^2$ fällt weg: $-4 = -2x - 16 \Rightarrow x = -6$
55. $-9x + 12 = -2x - 30 \Rightarrow x = 6$
56. $\frac{x}{5} = 3 \Rightarrow x = 15$
57. $x^2$ fällt weg: $6x + 8 = 7x + 12 \Rightarrow x = -4$
58. $6x + 4 = -4x - 36 \Rightarrow x = -4$
59. $\frac{x}{3} = -6 \Rightarrow x = -18$
60. $x^2$ fällt weg: $-4 = 9x - 49 \Rightarrow x = 5$
61. $5x - 5 = 4x - 10 \Rightarrow x = -5$
62. $\frac{x}{3} = 6 \Rightarrow x = 18$
63. $x^2$ fällt weg: $7x + 12 = 2x + 27 \Rightarrow x = 3$
64. $-10x - 10 = -x + 26 \Rightarrow x = -4$
65. $\frac{x}{3} = 1 \Rightarrow x = 3$
66. $x^2$ fällt weg: $7x + 10 = 6x + 15 \Rightarrow x =$
67. $-2x - 5 = 2x + 9 \Rightarrow x = -\frac{7}{2}$
68. $\frac{x}{2} = 6 \Rightarrow x = 12$
69. $x^2$ fällt weg: $2x - 3 = -3x + 2 \Rightarrow x =$
70. $3x - 3 = -2x + 2 \Rightarrow x = 1$
71. $\frac{x}{4} = 4 \Rightarrow x = 16$
72. $x^2$ fällt weg: $2x - 8 = -4 \Rightarrow x = 2$
73. $x + 2 = -4x + 7 \Rightarrow x = 1$
74. $\frac{x}{4} = 3 \Rightarrow x = 12$
75. $x^2$ fällt weg: $-2x - 8 = x - 23 \Rightarrow x =$
76. $-5x + 20 = 3x + 12 \Rightarrow x = 1$
77. $\frac{x}{5} = 5 \Rightarrow x = 25$
78. $x^2$ fällt weg: $4x + 3 = 7x - 9 \Rightarrow x = 4$
79. $4x - 4 = -2x + 14 \Rightarrow x = 3$
80. $\frac{x}{5} = -5 \Rightarrow x = -25$
81. $x^2$ fällt weg: $4x + 3 = -4x + 19 \Rightarrow x =$
82. $15x - 15 = -3x + 129 \Rightarrow x = 8$
83. $\frac{x}{5} = 5 \Rightarrow x = 25$
84. $x^2$ fällt weg: $-3x - 4 = 6x - 22 \Rightarrow x =$
85. $-6x - 2 = 4x - 27 \Rightarrow x = \frac{5}{2}$
86. $\frac{x}{4} = 2 \Rightarrow x = 8$
87. $x^2$ fällt weg: $4x + 3 = 3x + 5 \Rightarrow x = 2$
88. $5x + 10 = 2x + 22 \Rightarrow x = 4$
89. $x^2$ fällt weg: $3x - 4 = -2x - 34 \Rightarrow x =$
90. $-6x - 12 = 2x - 4 \Rightarrow x = -1$
91. $\frac{x}{2} = -5 \Rightarrow x = -10$
92. $x^2$ fällt weg: $-x - 2 = 5x - 32 \Rightarrow x =$
93. $-3x + 6 = 3x - 42 \Rightarrow x = 8$
94. $\frac{x}{4} = -3 \Rightarrow x = -12$
95. $x^2$ fällt weg: $4x + 3 = 2x + 1 \Rightarrow x = -$
96. $-9x + 3 = 2x + 36 \Rightarrow x = -3$
97. $\frac{x}{3} = 2 \Rightarrow x = 6$
98. $x^2$ fällt weg: $6x + 5 = x + 20 \Rightarrow x = 3$